

CATEGORIES OF BOOLEAN NETWORKS

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Table of Contents

Title Page	i
Abstract	ii
Dedication	ii
1 Introduction	1
2 Maps Between Boolean Networks	2
2.1 Conjugacy and isometric conjugacy	2
2.2 Semiconjugacy and contraction	11
3 Calculus of Boolean Networks	19
3.1 Morphisms and Boolean observables	19
3.2 Finite differences on the Boolean cube	21
3.3 Differentiation through a morphism	24
3.4 Wiring diagrams, paths, and feedback	27
3.5 Perturbations and iteration	30
3.6 Scalar factors and dynamical observables	32
3.7 Reconstruction and the information in a morphism	33
Concluding remarks	36
Bibliography	37

Chapter 1

Introduction

Chapter 2

Maps Between Boolean Networks

This chapter investigates how maps between Boolean networks can respect both their dynamics and the dependencies recorded by their wiring diagrams. Conjugate Boolean networks have isomorphic phase spaces, yet their wiring diagrams need not be isomorphic: an arbitrary relabeling of states need not preserve changes in a single coordinate. We therefore begin with conjugacies that preserve the Hamming distance on the Boolean cube. These isometric conjugacies induce isomorphisms of wiring diagrams and provide a natural notion of isomorphism for Boolean networks. We show that every isometry is a composition of a coordinate permutation and coordinate complementations. Such maps form the group of signed permutations, or the Weyl group of type B_n , isomorphic to the wreath product $C_2 \wr S_n$, where S_n is the symmetric group.

Our main goal is to extend this connection between dynamics and wiring diagrams to reductions that aggregate several nodes into one. A semiconjugacy is a surjective state map h satisfying $h \circ f = g \circ h$, so that the reduced network g describes the evolution of the observed states. Requiring h to be a contraction ensures that aggregation does not increase Hamming distances, but this alone does not ensure a corresponding relationship between wiring diagrams. We introduce a compatible node map ϕ , which partitions the source nodes into blocks and requires each coordinate of h to depend only on its assigned block. The main result, the edge-lifting theorem (Theorem 2.2.11), shows that under these conditions every edge $a \rightarrow b$ in the reduced network lifts to an edge $i \rightarrow j$ in the original network with $\phi(i) = a$ and $\phi(j) = b$. Thus every dependency observed after reduction has a witness among the original dependencies. Examples show why compatibility is needed and how aggregation can erase source dependencies. The theorem gives a precise structural guarantee for network reduction and extends the preservation of wiring diagrams under isometric conjugacy to maps that identify states and group nodes.

2.1 Conjugacy and isometric conjugacy

Definition 2.1.1. Let $f, g: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be Boolean networks. A conjugacy from f to g is a bijection $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ such that $h \circ f = g \circ h$. When such a map exists, we say that f and g are conjugate and write $f \sim g$.

We can represent the previous identity as the following commutative diagram.

$$\begin{array}{ccc}
\mathbb{F}_2^n & \xrightarrow{h} & \mathbb{F}_2^n \\
f \downarrow & & \downarrow g \\
\mathbb{F}_2^n & \xrightarrow{h} & \mathbb{F}_2^n
\end{array}$$

Consequently, one may either (i) update a state via f and subsequently relabel it through h , or (ii) first apply the relabeling h and then update via g ; in both cases the resulting state is identical. Furthermore, given a Boolean network f and a bijection h , there exists a unique map g that renders the corresponding diagram commutative, namely

$$g = h \circ f \circ h^{-1}.$$

Observe that the identity map conjugates f to itself. If $h \circ f = g \circ h$, then composing on the right with h^{-1} yields $f \circ h^{-1} = h^{-1} \circ g$, and hence h^{-1} conjugates g to f . Finally, if $h \circ f = g \circ h$ and $k \circ g = l \circ k$, then

$$(k \circ h) \circ f = k \circ (g \circ h) = l \circ (k \circ h),$$

which shows that $k \circ h$ conjugates f to l . Therefore,

Proposition 2.1.2. *Conjugacy is an equivalence relation on Boolean networks.*

Example 2.1.3. A paradigmatic instance of a structural correspondence is provided by the equivalence between OR-networks and AND-networks. Specifically, consider the maps $f, g: \mathbb{F}_2^3 \rightarrow \mathbb{F}_2^3$ defined by

$$f(x_1, x_2, x_3) = (x_1 x_2 x_3, x_1 x_2, x_1 x_3); \quad g(y_1, y_2, y_3) = (y_1 \vee y_2 \vee y_3, y_1 \vee y_2, y_1 \vee y_3).$$

The wiring diagrams associated with these two networks coincide; they are given below, where self-loops are understood and each edge is regarded as bidirectional:



Define $h: \mathbb{F}_2^3 \rightarrow \mathbb{F}_2^3$ by $h(x_1, x_2, x_3) = (\bar{x}_1, \bar{x}_2, \bar{x}_3)$. This map is a bijection (indeed, an involution). Moreover, a direct computation yields

$$\begin{aligned}
hf(x_1, x_2, x_3) &= h(x_1 x_2 x_3, x_1 x_2, x_1 x_3) \\
&= (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3, \bar{x}_1 \vee \bar{x}_2, \bar{x}_1 \vee \bar{x}_3) \\
&= g(\bar{x}_1, \bar{x}_2, \bar{x}_3) \\
&= gh(x_1, x_2, x_3),
\end{aligned}$$

and hence $hf = gh$, establishing that f and g are conjugate via h .

It should be observed that the conjugacy identity is invariant under arbitrary iteration. More precisely, for every $k \in \mathbb{N}$,

$$h \circ f^k = g^k \circ h.$$

Here, f^k (respectively, g^k) denotes the k -fold iterate of f (respectively, g), i.e., the k -fold self-

composition. It follows that h maps the entire forward orbit of x ,

$$x, f(x), f^2(x), \dots,$$

onto the forward orbit of $h(x)$,

$$h(x), g(h(x)), g^2(h(x)), \dots,$$

in a term-wise manner. Equivalently, in graph-theoretic language, a conjugacy is precisely an isomorphism of directed graphs between the corresponding phase spaces.

Conjugacy exhibits numerous desirable properties that are compatible with dynamical behavior; we have mentioned only a subset of these. Many such properties are investigated in [5, 6, 8, 4], among other works. However, some limitations have been found regarding the geometry of the Boolean networks; we will see this explicitly in the following example.

Example 2.1.4. Let $f, g: \mathbb{F}_2^3 \rightarrow \mathbb{F}_2^3$ be the Boolean networks defined by

$$f(x_0, x_1, x_2) = (x_0, 0, 0)$$

and

$$g(x_0, x_1, x_2) = (p(x), p(x), p(x)), \quad p(x) = x_0 + x_1 + x_2,$$

where $x = (x_0, x_1, x_2)$ and $+$ denotes addition modulo 2. Define a map $\pi: \mathbb{F}_2^3 \rightarrow \mathbb{F}_2^3$ by

$$\pi(x_0, x_1, x_2) = (x_0 + x_1, x_0 + x_2, x_0 + x_1 + x_2).$$

To verify that π is bijective, define

$$\pi^{-1}(y_0, y_1, y_2) = (y_0 + y_1 + y_2, y_1 + y_2, y_0 + y_2).$$

Using the identities $a + a = 0$ and $a + 0 = a$, together with the associativity and commutativity of $+$, we obtain

$$\begin{aligned} \pi^{-1}(\pi(x_0, x_1, x_2)) &= ((x_0 + x_1) + (x_0 + x_2) + (x_0 + x_1 + x_2), \\ &\quad (x_0 + x_2) + (x_0 + x_1 + x_2), \\ &\quad (x_0 + x_1) + (x_0 + x_1 + x_2)) \\ &= (x_0, x_1, x_2). \end{aligned}$$

Conversely, if

$$(x_0, x_1, x_2) = \pi^{-1}(y_0, y_1, y_2) = (y_0 + y_1 + y_2, y_1 + y_2, y_0 + y_2),$$

then

$$\begin{aligned} \pi(\pi^{-1}(y_0, y_1, y_2)) &= ((y_0 + y_1 + y_2) + (y_1 + y_2), \\ &\quad (y_0 + y_1 + y_2) + (y_0 + y_2), \\ &\quad (y_0 + y_1 + y_2) + (y_1 + y_2) + (y_0 + y_2)) \\ &= (y_0, y_1, y_2). \end{aligned}$$

Thus π^{-1} is the inverse of π , so π is a permutation of the eight-element set $\mathbb{F}_2^3$.

For every $(x_0, x_1, x_2) \in \mathbb{F}_2^3$,

$$\pi(f(x_0, x_1, x_2)) = \pi(x_0, 0, 0) = (x_0, x_0, x_0).$$

Now let $(y_0, y_1, y_2) = \pi(x_0, x_1, x_2)$. Then

$$\begin{aligned} y_0 + y_1 + y_2 &= (x_0 + x_1) + (x_0 + x_2) + (x_0 + x_1 + x_2) \\ &= x_0. \end{aligned}$$

It follows that

$$g(\pi(x_0, x_1, x_2)) = g(y_0, y_1, y_2) = (x_0, x_0, x_0).$$

Therefore,

$$g \circ \pi = \pi \circ f.$$

The corresponding wiring diagrams of f and g are as follows:



The only essential dependence in f is that of f_0 on x_0 , whereas each coordinate function of g depends essentially on all three variables. Hence,

$$|E(W(f))| = 1, \quad |E(W(g))| = 9.$$

Since an isomorphism of directed graphs preserves the number of edges,

$$W(f) \not\cong W(g).$$

A general theory of the above example is explored in [7].

The preceding example shows that the phase space alone does not determine the wiring diagram. The distinction arises because a wiring diagram records the effect of changing one coordinate at a time, whereas an arbitrary conjugacy need not preserve such changes. In fact, the map π in Example 2.1.4 sends the states $(0, 0, 0)$ and $(1, 0, 0)$, which differ in only one coordinate, to $(0, 0, 0)$ and $(1, 1, 1)$, which differ in all three coordinates. To obtain a notion of conjugacy that also preserves the wiring diagram, we therefore seek relabelings that respect the geometry of the Boolean cube.

2.1.1 The Hamming distance and the Boolean cube

Definition 2.1.5. For $x, y \in \mathbb{F}_2^n$, the *Hamming distance* between x and y is

$$d_H(x, y) = |\{i \in \{1, \dots, n\} : x_i \neq y_i\}|.$$

The *Boolean cube* Q_n is the undirected graph with vertex set $\mathbb{F}_2^n$, in which two states are adjacent if their Hamming distance is 1.

The Hamming distance is a metric. Indeed, it is nonnegative and symmetric, and $d_H(x, y) = 0$ if and only if $x = y$. For the triangle inequality, observe that if $x_i \neq z_i$, then either $x_i \neq y_i$ or $y_i \neq z_i$. Counting these coordinates gives

$$d_H(x, z) \leq d_H(x, y) + d_H(y, z).$$

Moreover, $d_H(x, y)$ is precisely the length of a shortest path from x to y in Q_n : every differing coordinate must be changed at least once, and changing each of them exactly once provides such a path. Thus Q_n is the graph of vertices and edges of the unit cube $[0, 1]^n$. Its edges describe single-coordinate changes, independently of any Boolean network defined on its vertices.

Example 2.1.6. In $\mathbb{F}_2^3$, we have

$$d_H((0, 0, 0), (1, 0, 0)) = 1, \quad d_H((0, 0, 0), (1, 1, 1)) = 3.$$

Returning to the conjugacy π in Example 2.1.4, these equalities show that

$$d_H(\pi(0, 0, 0), \pi(1, 0, 0)) \neq d_H((0, 0, 0), (1, 0, 0)).$$

The highlighted edge on the left is sent to the pair of opposite vertices on the right. The dashed segment is a body diagonal of the unit cube, not an edge of Q_3 .



Here and in subsequent cube drawings, a string such as 100 denotes the state $(1, 0, 0)$. The two cubes above display the selected pair and its image under π ; their remaining vertices retain their usual coordinate labels.

Definition 2.1.7. A map $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is an *isometry* of the Boolean cube if

$$d_H(h(x), h(y)) = d_H(x, y) \quad \text{for all } x, y \in \mathbb{F}_2^n.$$

A Boolean network is called *isometric* if its update map is an isometry.

Every isometry is injective, since $h(x) = h(y)$ implies $d_H(x, y) = 0$, and hence $x = y$. Since $\mathbb{F}_2^n$ is finite, every isometry of the Boolean cube is therefore bijective. Its inverse is also an isometry. Equivalently, an isometry is an automorphism of Q_n : distance preservation gives adjacency preservation in both directions, and a graph automorphism preserves shortest-path lengths.

Definition 2.1.8. Let $f, g: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be Boolean networks. An *isometric conjugacy* from f to g is an isometry $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ such that

$$h \circ f = g \circ h.$$

When such a map exists, we say that f and g are *isometrically conjugate* and write $f \sim_{\text{iso}} g$.

In this definition, the distance-preserving condition is imposed on h ; the update maps f and g need not themselves be isometries. Since the identity, inverses, and compositions of isometries are isometries, the same argument used for conjugacy shows that isometric conjugacy is an equivalence relation on Boolean networks.

Example 2.1.9. Consider the map $h: \mathbb{F}_2^3 \rightarrow \mathbb{F}_2^3$ defined by

$$h(x_1, x_2, x_3) = (\bar{x}_3, x_1, \bar{x}_2).$$

Complementing a coordinate does not change whether two entries agree, and permuting the coordinates does not change the number of disagreements. Consequently,

$$\begin{aligned} d_H(h(x), h(y)) &= |\bar{x}_3 - \bar{y}_3| + |x_1 - y_1| + |\bar{x}_2 - \bar{y}_2| \\ &= |x_3 - y_3| + |x_1 - y_1| + |x_2 - y_2| \\ &= d_H(x, y), \end{aligned}$$

where the differences in this calculation are taken in $\mathbb{R}$. Thus h is an isometry, with inverse

$$h^{-1}(y_1, y_2, y_3) = (y_2, \bar{y}_3, \bar{y}_1).$$

The following drawings show the action of h on every vertex of the unit cube. Vertices in corresponding positions are related by h . The blue, orange, and gray edges represent changes in x_1, x_2, x_3 , respectively; their images represent changes in y_2, y_3, y_1 .



The complementation map from Example 2.1.3 is also an isometry. Hence the AND-network and OR-network in that example are isometrically conjugate. Neither update map is an isometry: the AND-network sends both $(0, 0, 0)$ and $(0, 0, 1)$ to $(0, 0, 0)$, while the OR-network sends both $(1, 0, 0)$ and $(1, 1, 0)$ to $(1, 1, 1)$.

2.1.2 Preservation of the wiring diagram

We now show that isometric conjugacy preserves the wiring diagram up to a relabeling of its vertices. Let $e_i \in \mathbb{F}_2^n$ denote the vector whose i th coordinate is 1 and whose other coordinates are 0. We use $+$ componentwise on vectors, so $x + e_i$ is the state obtained from x by changing its i th coordinate. Recall that the wiring diagram $W(f)$ has vertex set $\{1, \dots, n\}$, with

$$i \longrightarrow j \text{ in } W(f) \iff f_j(x) \neq f_j(x + e_i) \text{ for some } x \in \mathbb{F}_2^n. \quad (2.1)$$

This is the unsigned wiring diagram; self-loops are included. Its vertices are coordinates, whereas the vertices of Q_n are states.

The essential geometric point is that an isometry sends all edges in a given coordinate direction to edges in one fixed coordinate direction. The choice of the new direction cannot depend on the state.

Lemma 2.1.10. *Let $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be an isometry. There exists a unique permutation σ of $\{1, \dots, n\}$ such that*

$$h(x + e_i) = h(x) + e_{\sigma(i)} \quad \text{for every } x \in \mathbb{F}_2^n \text{ and every } i. \quad (2.2)$$

Moreover, for all $u, v \in \mathbb{F}_2^n$,

$$h(u)_{\sigma(j)} \neq h(v)_{\sigma(j)} \iff u_j \neq v_j. \quad (2.3)$$

Proof. Since h is an automorphism of Q_n , it maps the n neighbors of each x bijectively onto the n neighbors of $h(x)$. Thus there is a permutation σ_x such that

$$h(x + e_i) = h(x) + e_{\sigma_x(i)}.$$

Traversing the same edge in the opposite direction shows that $\sigma_{x+e_i}(i) = \sigma_x(i)$.

Now let $i \neq j$. The four distinct states

$$x, \quad x + e_i, \quad x + e_i + e_j, \quad x + e_j$$

form a square in Q_n . Put $z = h(x)$, $r = \sigma_x(i)$, and $s = \sigma_x(j)$. The two common neighbors of $z + e_r$ and $z + e_s$ are precisely z and $z + e_r + e_s$. Since h is injective, the vertex opposite x must therefore satisfy

$$h(x + e_i + e_j) = z + e_r + e_s = h(x + e_j) + e_r.$$

Therefore, $\sigma_{x+e_j}(i) = \sigma_x(i)$. Consequently, $\sigma_x(i)$ is unchanged when any coordinate of x is changed. The cube is connected, so σ_x is a single permutation σ , independent of x . Its uniqueness follows from the images of the edges incident with $\mathbf{0}$.

Finally, join u to v by changing each coordinate in which they differ exactly once. Equation (2.2) shows that the image path changes precisely the corresponding coordinates $\sigma(i)$, again exactly once. In particular, its endpoints differ in coordinate $\sigma(j)$ if and only if u and v differ in coordinate j , proving equation (2.3). $\square$

Proposition 2.1.11. *Let $f, g: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be isometrically conjugate via h , and let σ be the permutation associated with h in Lemma 2.1.10. Then*

$$i \longrightarrow j \text{ in } W(f) \iff \sigma(i) \longrightarrow \sigma(j) \text{ in } W(g).$$

Thus $\sigma: W(f) \rightarrow W(g)$ is an isomorphism of directed graphs.

Proof. Fix $x \in \mathbb{F}_2^n$ and put $y = h(x)$. By equation (2.2),

$$h(x + e_i) = y + e_{\sigma(i)}.$$

The conjugacy identity gives

$$g(y) = h(f(x)), \quad g(y + e_{\sigma(i)}) = h(f(x + e_i)).$$

Applying equation (2.3) to these two output states yields

$$g_{\sigma(j)}(y) \neq g_{\sigma(j)}(y + e_{\sigma(i)}) \iff f_j(x) \neq f_j(x + e_i).$$

Since h is bijective, y ranges over all of $\mathbb{F}_2^n$ as x does. Taking the existence of such a state on both sides and using equation (2.1) proves the assertion, including the case $i = j$. $\square$

Example 2.1.12. Consider the Boolean network

$$f(x_1, x_2, x_3) = (x_1 x_2, x_1 \vee x_3, x_1 + x_2)$$

and the isometry $h(x_1, x_2, x_3) = (\bar{x}_3, x_1, \bar{x}_2)$ from Example 2.1.9. Define $g = h \circ f \circ h^{-1}$. Substituting

the expression for h^{-1} gives

$$\begin{aligned} g(y_1, y_2, y_3) &= h(y_2\bar{y}_3, y_2 \vee \bar{y}_1, y_2 + \bar{y}_3) \\ &= (y_2 + y_3, y_2\bar{y}_3, y_1\bar{y}_2). \end{aligned}$$

For example, at $x = (1, 0, 1)$,

$$f(x) = (0, 1, 1), \quad h(f(x)) = (0, 0, 0),$$

while $h(x) = (0, 1, 1)$ and $g(h(x)) = (0, 0, 0)$, as required by the conjugacy identity.

The coordinate directions are permuted according to

$$\sigma(1) = 2, \quad \sigma(2) = 3, \quad \sigma(3) = 1.$$

The wiring diagrams are drawn below, with corresponding vertices placed in corresponding positions. Every arrow on the left has its image on the right; in particular, the self-loop at 1 becomes the self-loop at 2.



The corresponding map of unit cubes is exactly the one displayed in Example 2.1.9. The wiring diagrams have only three vertices, one for each coordinate, while each cube has eight vertices, one for each state. In this example, $W(f)$ and $W(g)$ are isomorphic but are not equal as labeled graphs: their self-loops have different labels.

Remark 2.1.13. Proposition 2.1.11 concerns unsigned wiring diagrams. Complementing coordinates can change whether a dependence is increasing or decreasing. For example, the first coordinate function $f_1 = x_1x_2$ in Example 2.1.12 is increasing in x_2 , whereas the corresponding function $g_2 = y_2\bar{y}_3$ is decreasing in y_3 . Thus preservation of the unsigned diagram does not assert preservation of individual edge signs.

2.1.3 Characterization of the isometries

Example 2.1.9 suggests that coordinate permutations and coordinate complementations are the basic isometries of the Boolean cube. We now show that every isometry is obtained in this way. This standard characterization is also recorded in [2].

For a permutation σ of $\{1, \dots, n\}$, let $P_\sigma: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be the coordinate permutation defined by

$$(P_\sigma(x))_{\sigma(i)} = x_i, \quad \text{equivalently} \quad (P_\sigma(x))_j = x_{\sigma^{-1}(j)}.$$

Thus an input coordinate i is moved to output coordinate $\sigma(i)$, consistently with the relabeling of the wiring diagram above. For $a \in \mathbb{F}_2^n$, define $\tau_a(x) = a + x$. This map complements precisely those coordinates j for which $a_j = 1$.

Theorem 2.1.14. *A map $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is an isometry if and only if there exist $a \in \mathbb{F}_2^n$ and a permutation σ of $\{1, \dots, n\}$ such that*

$$h = \tau_a \circ P_\sigma.$$

The pair (a, σ) is unique, with $a = h(\mathbf{0})$ and

$$h(e_i) + h(\mathbf{0}) = e_{\sigma(i)}.$$

Proof. Suppose first that h is an isometry. Put $a = h(\mathbf{0})$ and let σ be the permutation from Lemma 2.1.10. Starting at $\mathbf{0}$ and changing each coordinate i for which $x_i = 1$, equation (2.2) gives

$$h(x) = a + \sum_{\{i: x_i=1\}} e_{\sigma(i)} = a + P_\sigma(x).$$

Here the empty sum is $\mathbf{0}$. Evaluating at $\mathbf{0}$ determines a , and evaluating at each e_i then determines σ , proving uniqueness.

Conversely, suppose that $h(x) = a + P_\sigma(x)$. For every $x, y \in \mathbb{F}_2^n$,

$$\begin{aligned} d_H(h(x), h(y)) &= |\{j : a_j + x_{\sigma^{-1}(j)} \neq a_j + y_{\sigma^{-1}(j)}\}| \\ &= |\{j : x_{\sigma^{-1}(j)} \neq y_{\sigma^{-1}(j)}\}| \\ &= d_H(x, y). \end{aligned}$$

Hence h is an isometry. □

Consequently, there are exactly $2^n n!$ isometries of Q_n , and an isometry is completely determined by its values at $\mathbf{0}, e_1, \dots, e_n$. In particular, an isometry fixing $\mathbf{0}$ is a coordinate permutation. The characterization also gives the coordinate form of an isometric conjugacy: if $g = h \circ f \circ h^{-1}$ and $h = \tau_a \circ P_\sigma$, then

$$g_j(y) = a_j + f_{\sigma^{-1}(j)}(P_{\sigma^{-1}}(y + a)).$$

When σ is the identity, the wiring diagram is unchanged as a labeled graph. For a general σ , its vertices are relabeled by σ .

Example 2.1.15. For $h(x_1, x_2, x_3) = (\bar{x}_3, x_1, \bar{x}_2)$, the images of the origin and its three neighbors are

$$h(000) = 101, \quad h(100) = 111, \quad h(010) = 100, \quad h(001) = 001.$$

Thus $a = (1, 0, 1)$, and

$$h(e_1) + a = e_2, \quad h(e_2) + a = e_3, \quad h(e_3) + a = e_1.$$

We recover $\sigma = (123)$ and

$$h(x) = (1, 0, 1) + (x_3, x_1, x_2),$$

in agreement with the two labeled cubes in Example 2.1.9. These four images determine the images of all eight states. By contrast, the map π from Example 2.1.4 fixes 000 but sends 100 to 111; it therefore cannot be a coordinate permutation and hence cannot be an isometry. Proposition 2.1.11 further shows that no other conjugacy between the two networks in Example 2.1.4 can be isometric, since their wiring diagrams are not isomorphic.

The characterization has a particularly simple interpretation when the update map of a Boolean network is itself an isometry.

Corollary 2.1.16. *A Boolean network $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is isometric if and only if every vertex of $W(f)$ has exactly one incoming edge and exactly one outgoing edge. Equivalently, $W(f)$ is a disjoint union of directed cycles, with self-loops allowed as cycles of length one.*

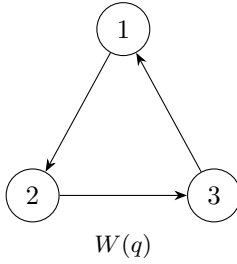
Proof. If f is isometric, Theorem 2.1.14 gives $f_{\sigma(i)}(x) = a_{\sigma(i)} + x_i$. Its wiring diagram therefore has exactly the edges $i \rightarrow \sigma(i)$, so every vertex has one incoming and one outgoing edge.

Conversely, suppose that every vertex has one incoming and one outgoing edge, and write its unique outgoing edge as $i \rightarrow \sigma(i)$. The incoming-edge condition makes σ a permutation. The function $f_{\sigma(i)}$ depends essentially only on x_i : changing any other coordinate does not change its value, and any two states with the same i th coordinate can be joined by such changes. A nonconstant Boolean function of one variable is either that variable or its complement. Hence $f_{\sigma(i)}(x) = a_{\sigma(i)} + x_i$ for some $a_{\sigma(i)} \in \mathbb{F}_2$, and Theorem 2.1.14 applies. $\square$

Example 2.1.17. Regard the isometry from Example 2.1.9 as an update map, and denote the resulting network by

$$q(x_1, x_2, x_3) = (\bar{x}_3, x_1, \bar{x}_2).$$

Its wiring diagram and all of its periodic orbits are given below.



Fixed points: 001 and 110.

$$000 \rightarrow 101 \rightarrow 011 \rightarrow 000,$$

$$100 \rightarrow 111 \rightarrow 010 \rightarrow 100.$$

The corresponding cube map appears in Example 2.1.9. The directed cycle in $W(q)$ records coordinate dependencies, whereas the two cycles of length three above record the evolution of states. Although q preserves distances between pairs of updated states, it need not move each state along a cube edge: $d_H(000, q(000)) = 2$.

2.2 Semiconjugacy and contraction

Conjugacy describes a change of labels on the state space. To compare a network with a smaller system that records only some of its information, we must also allow distinct states to have the same image. We therefore replace the bijective state map by a surjective one. The dynamical condition remains a commutative square, but its consequences for the Boolean cube and the wiring diagram require separate consideration.

Throughout this section, $n, m \geq 1$, $[n] = \{1, \dots, n\}$, and $+$ denotes addition modulo 2. We use the Hamming distance on cubes of either dimension, and e_i denotes the standard basis vector in the cube indicated by its context.

2.2.1 Semiconjugacy and the identification of states

Definition 2.2.1. Let $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ and $g: \mathbb{F}_2^m \rightarrow \mathbb{F}_2^m$ be Boolean networks. A *semiconjugacy* from f to g is a surjective Boolean map $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ such that

$$h \circ f = g \circ h. \tag{2.4}$$

Thus the following diagram commutes:

$$\begin{array}{ccc} \mathbb{F}_2^n & \xrightarrow{h} & \mathbb{F}_2^m \\ f \downarrow & & \downarrow g \\ \mathbb{F}_2^n & \xrightarrow{h} & \mathbb{F}_2^m \end{array}$$

When such a map exists, g is called a *factor* of f . Surjectivity is part of our convention for semiconjugacy.

The map h records an observation of a state. Equation (2.4) says that this observation can be updated using g without knowing which state in the fiber $h^{-1}(y)$ produced it. Surjectivity ensures that every state of the smaller network is represented. It also implies $m \leq n$. If $m = n$, then h is bijective and the semiconjugacy is a conjugacy.

As with conjugacy, induction gives $h \circ f^k = g^k \circ h$ for every $k \geq 0$. Thus h maps each forward orbit onto the corresponding forward orbit in the factor. Distinct trajectories can merge under h , and the period of a periodic state can decrease to a divisor of its original period. Identity maps are semiconjugacies, and compositions of semiconjugacies are semiconjugacies.

Proposition 2.2.2. *Let $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ and let $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ be surjective. There is a Boolean network g satisfying equation (2.4) if and only if*

$$h(x) = h(x') \implies h(f(x)) = h(f(x')) \quad (x, x' \in \mathbb{F}_2^n). \quad (2.5)$$

If g exists, it is unique.

Proof. Necessity follows by applying g to $h(x) = h(x')$. Conversely, for $y \in \mathbb{F}_2^m$, choose $x \in h^{-1}(y)$ and put $g(y) = h(f(x))$. The condition in equation (2.5) makes this definition independent of the choice of x . It gives $gh = hf$, and surjectivity of h forces these values of g at every y . $\square$

Example 2.2.3. Consider the eight-node network

$$f(x) = (x_2, x_3, x_4, \bar{x}_1, x_1 + x_5, x_2x_6, x_3 \vee x_7, \bar{x}_8)$$

and the four-node network

$$g(y) = (y_2, y_3, y_4, \bar{y}_1).$$

The projection $h(x) = (x_1, x_2, x_3, x_4)$ is surjective, and

$$h(f(x)) = (x_2, x_3, x_4, \bar{x}_1) = g(h(x)).$$

Every state y has 16 preimages: its first four coordinates are prescribed, and the last four are arbitrary. Although the discarded coordinates have their own dynamics and receive inputs from retained coordinates, they do not affect the evolution of the observation.

This reduction cannot preserve all distances. For instance, $h(\mathbf{0}) = h(e_5)$ even though $d_H(\mathbf{0}, e_5) = 1$. It does satisfy

$$d_H(h(x), h(x')) = |\{i \leq 4 : x_i \neq x'_i\}| \leq d_H(x, x').$$

Thus information can be discarded without increasing the size of a perturbation.

Example 2.2.4. Semiconjugacy alone does not impose the preceding inequality. In fact, its failure can be arbitrarily large. Let $m \geq 2$ and $r \geq 1$, write a source state as $(u, z) \in \mathbb{F}_2^m \times \mathbb{F}_2^r$, and let

$R: \mathbb{F}_2^r \rightarrow \mathbb{F}_2^r$ cyclically permute the coordinates. Define

$$\begin{aligned} h(u, z) &= (u_1, u_1 + u_2, \dots, u_1 + u_m), \\ f(u, z) &= (\bar{u}_1, 0, \dots, 0; R(z)), \\ g(y) &= (\bar{y}_1, \bar{y}_1, \dots, \bar{y}_1). \end{aligned}$$

Here the semicolon separates the first m updated coordinates from the last r . Given y , take $u_1 = y_1$ and $u_i = y_1 + y_i$ for $2 \leq i \leq m$; any z then gives a preimage. Moreover,

$$h(f(u, z)) = (\bar{u}_1, \dots, \bar{u}_1) = g(h(u, z)).$$

These are nonconstant dynamics, and g exchanges $\mathbf{0}$ and $\mathbf{1}$. Nevertheless,

$$d_H(\mathbf{0}, e_1) = 1, \quad d_H(h(\mathbf{0}), h(e_1)) = m.$$

One changed input can therefore appear as a simultaneous change in every observed coordinate. The case $m = 5$, $r = 3$ already gives an eight-node network mapping onto a five-node network with this behavior.

2.2.2 Contractions of Boolean cubes

Distance preservation would force h to be injective, excluding every proper reduction of the state space. Example 2.2.3 suggests allowing distances to decrease, while Example 2.2.4 explains why an upper bound is an additional condition.

Definition 2.2.5. A Boolean map $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is a *contraction* if

$$d_H(h(x), h(x')) \leq d_H(x, x') \quad \text{for all } x, x' \in \mathbb{F}_2^n. \quad (2.6)$$

A semiconjugacy satisfying this condition is a *contractive semiconjugacy*.

The inequality in equation (2.6) is usually called *Hamming nonexpansiveness*, or the 1-Lipschitz condition; see, for example, [9]. We use *contraction* for this non-strict inequality. A bound with a constant strictly smaller than 1 would identify the endpoints of every cube edge, and connectedness of the cube would force the map to be constant. The contraction condition here concerns h ; the update maps f and g need not be contractions.

Lemma 2.2.6. A map $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is a contraction if and only if

$$d_H(h(x + e_i), h(x)) \leq 1 \quad \text{for every } x \in \mathbb{F}_2^n \text{ and } i \in [n]. \quad (2.7)$$

Equivalently, each edge of Q_n maps either to an edge of Q_m or to a single vertex.

Proof. Necessity follows by taking two inputs at distance 1. Conversely, join x to x' by a path of length $d_H(x, x')$ in Q_n . Each step changes the image by distance at most 1. The triangle inequality bounds the distance between the image endpoints by the length of this path. $\square$

For Boolean self-maps, this criterion says that changing one input affects at most one output at any fixed state. It is the local out-degree characterization of nonexpansiveness in [9, Proposition 2]. The proof above applies also when the two cubes have different dimensions. Compositions of contractions are contractions. A surjective contraction from $\mathbb{F}_2^n$ to itself is an isometry. Indeed, it is a bijection of a finite set, so $h^N = \text{Id}$ for some $N \geq 1$. Repeated application of equation (2.6) gives

$$d_H(x, x') \geq d_H(h(x), h(x')) \geq \dots \geq d_H(h^N(x), h^N(x')) = d_H(x, x'),$$

so all inequalities are equalities.

The edge criterion leaves open which output coordinate changes. Unlike Lemma 2.1.10, it does not assert that this coordinate is determined by the input direction alone. The following example shows both this obstruction and its effect on wiring diagrams.

Example 2.2.7. For $u \in \mathbb{F}_2^3$, let $|u|$ be its number of nonzero coordinates, and define $H: \mathbb{F}_2^3 \rightarrow \mathbb{F}_2^2$ by

$$\begin{array}{c|cccc} |u| & 0 & 1 & 2 & 3 \\ \hline H(u) & 00 & 10 & 11 & 01 \end{array}.$$

The four possible outputs occur, so H is surjective. Adjacent input states have consecutive weights, and consecutive entries in the second row are adjacent in Q_2 . Lemma 2.2.6 therefore shows that H is a contraction.

Consider the networks

$$F(u_1, u_2, u_3) = (\bar{u}_3, \bar{u}_1, \bar{u}_2), \quad G(v_1, v_2) = (v_1, \bar{v}_2).$$

Since $|F(u)| = 3 - |u|$, the table gives $H(F(u)) = G(H(u))$. Thus H is a contractive semiconjugacy between two nonidentity networks. Nevertheless, two edges in input direction 1 have images in different directions:

$$\begin{array}{ccc} 000 \longleftrightarrow 100 & \xrightarrow{H} & 00 \longleftrightarrow 10, \\ 010 \longleftrightarrow 110 & \xrightarrow{H} & 10 \longleftrightarrow 11. \end{array}$$

Hence no single output direction can be assigned to this input direction at every state.

There is also a direct obstruction to lifting wiring edges. The diagrams are



Every surjection $[3] \rightarrow [2]$ has a singleton fiber. The loop at its image node would have to lift to a loop at that single source node, but $W(F)$ has no loops. Thus contraction and semiconjugacy do not even guarantee the existence of a surjective node map lifting every target edge.

This obstruction persists in larger systems with interacting additional coordinates. For $r \geq 2$, let R cyclically permute r coordinates and set

$$h(u, z) = (H(u), z), \quad f(u, z) = (F(u), R(z)), \quad g(v, z) = (G(v), R(z)).$$

These give a contractive semiconjugacy from $r + 3$ nodes to $r + 2$ nodes. The source wiring diagram is the disjoint union of a directed 3-cycle and a directed r -cycle, so it has no loops. The target has two loops and a directed r -cycle. Any surjection $[r + 3] \rightarrow [r + 2]$ has exactly one fiber of size 2 and all remaining fibers of size 1. Both loop vertices would need a fiber of size at least 2, which is impossible. Taking $r = 4$ gives a seven-node to six-node example, and r can be arbitrarily large.

2.2.3 Compatible node maps and aggregation

To connect a state map to wiring diagrams, we need a consistent assignment of input coordinates to output coordinates. We allow an input perturbation to disappear, and we allow

several inputs to affect the same output. We require that an input cannot affect different outputs at different states.

For example, consider

$$h(x_1, \dots, x_9) = (x_1 \vee x_2 \vee x_3, x_4 x_5 x_6, x_7 + x_8 + x_9).$$

Changing any of the first three inputs affects only the first output, and the other two groups behave similarly. Some changes disappear: changing x_1 has no effect when $x_2 = 1$. Nevertheless, there is an unambiguous assignment of all nine source nodes to three target nodes. Each aggregation function is nonconstant, and its inputs are disjoint from the inputs of the others, so every output state can be attained independently. This example suggests the following definition.

Definition 2.2.8. Let $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ be a surjective Boolean map. A *compatible node map* for h is a surjection $\phi: [n] \rightarrow [m]$ such that

$$h(x + e_i) + h(x) \in \{\mathbf{0}, e_{\phi(i)}\} \quad \text{for every } x \in \mathbb{F}_2^n \text{ and } i \in [n]. \quad (2.8)$$

We write $B_a = \phi^{-1}(a)$ for its *node fibers*. Thus the sets B_a form a partition of $[n]$ into nonempty blocks.

The maps $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ and $\phi: [n] \rightarrow [m]$ act on different sets: h identifies states, while ϕ groups nodes. In particular, a node fiber B_a is not a state fiber $h^{-1}(y)$. No condition on the edges of $W(f)$ or $W(g)$ is included in Definition 2.2.8; the edge-lifting conclusion will be deduced from compatibility and semiconjugacy.

Proposition 2.2.9. Let $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ be surjective and let $\phi: [n] \rightarrow [m]$ be a surjection. The following conditions are equivalent:

- (i) ϕ is compatible with h .
- (ii) For every $a \in [m]$, the function h_a depends only on the coordinates in $B_a = \phi^{-1}(a)$.
- (iii) There are nonconstant Boolean functions $H_a: \mathbb{F}_2^{B_a} \rightarrow \mathbb{F}_2$ such that

$$h(x) = (H_1(x|_{B_1}), \dots, H_m(x|_{B_m})). \quad (2.9)$$

Here $\mathbb{F}_2^{B_a}$ denotes states indexed by the nodes in B_a , and $x|_{B_a}$ is the restriction of x to those coordinates. These conditions imply that h is a contraction.

Proof. Suppose (i) holds. If $i \notin B_a$, equation (2.8) gives $h_a(x + e_i) = h_a(x)$. Any two states agreeing on B_a can be joined by changing only coordinates outside B_a , so they have the same value of h_a . This proves (ii). Condition (ii) defines H_a by restriction, and surjectivity of h ensures that each h_a , hence each H_a , takes both Boolean values. Thus (iii) holds.

Conversely, under (iii), changing an input coordinate i can affect only the output with $a = \phi(i)$, which gives (i). Finally, equation (2.8) implies equation (2.7), so Lemma 2.2.6 gives contraction. $\square$

This proposition identifies the extra restriction beyond contraction. At a fixed state, a contraction permits an input to affect at most one output; compatibility requires the same assignment at every state. Equivalently, a surjective h admits a compatible node map precisely when each input coordinate is essential in at most one of the functions $h_1, \dots, h_m$. An input essential in h_a must be assigned to a . Inputs ignored by every component of h can be assigned to any node. Each output has an essential input because it is nonconstant, so these assignments give a surjection. When a compatible node map exists, it is unique exactly when every input is used or $m = 1$.

The projection in Example 2.2.3 therefore has compatible node maps: the first four nodes have forced assignments, while the last four can be assigned freely. The map in Example 2.2.7 has none. In the isometric case, the node fibers are singletons and the compatible map is precisely the permutation in Lemma 2.1.10.

Example 2.2.10. We now construct a semiconjugacy from a twelve-node network to a four-node network, using four different aggregation rules. Partition [12] into

$$B_1 = \{1, 2, 3\}, \quad B_2 = \{4, 5, 6\}, \quad B_3 = \{7, 8, 9\}, \quad B_4 = \{10, 11, 12\},$$

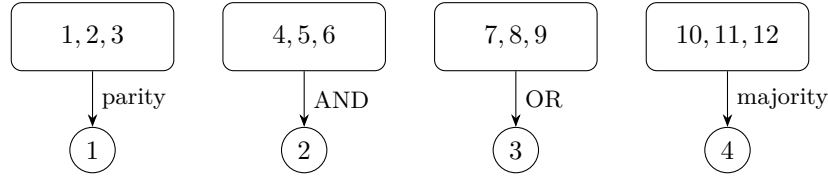
and put

$$\begin{aligned} P(x) &= x_1 + x_2 + x_3, & A(x) &= x_4 x_5 x_6, \\ O(x) &= x_7 \vee x_8 \vee x_9, & M(x) &= (x_{10} x_{11}) \vee (x_{10} x_{12}) \vee (x_{11} x_{12}). \end{aligned}$$

Thus M is the majority function on the last three inputs. Set

$$h(x) = (P(x), A(x), O(x), M(x)), \quad \phi(i) = a \quad \text{for } i \in B_a.$$

The assignment of nodes is displayed below; the arrows describe ϕ , rather than wiring edges.



Each rule takes a constant triple (t, t, t) to t , so

$$s(y) = (y_1, y_1, y_1, y_2, y_2, y_2, y_3, y_3, y_3, y_4, y_4, y_4)$$

satisfies $h(s(y)) = y$. Hence h is surjective, and Proposition 2.2.9 shows that it is a contraction with compatible node map ϕ . Every source coordinate is essential in its aggregation rule, so ϕ is unique. This map reduces 4096 states to 16 states. Its state fibers need not have equal sizes: the fibers over $(0, 1, 0, 0)$ and $(0, 0, 1, 0)$ have sizes $4 \cdot 1 \cdot 1 \cdot 4 = 16$ and $4 \cdot 7 \cdot 7 \cdot 4 = 784$, respectively.

Define the target network by

$$g(y) = (y_1 y_2 + y_3, y_1 \vee y_4, y_2 + y_4, y_1(y_3 \vee y_4)). \quad (2.10)$$

For the construction of f , abbreviate the four prescribed aggregate updates as

$$u_1 = PA + O, \quad u_2 = P \vee M, \quad u_3 = A + M, \quad u_4 = P(O \vee M).$$

These are explicit functions of x , namely the components of $g(h(x))$. Define f block by block:

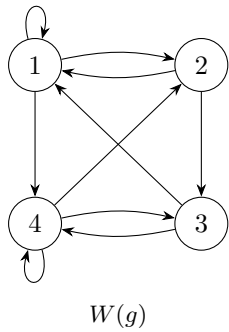
$$\begin{aligned} (f_1, f_2, f_3) &= (x_{10}, x_{11}, u_1 + x_{10} + x_{11}), \\ (f_4, f_5, f_6) &= (u_2 \vee x_7, u_2 \vee \bar{x}_7, u_2 \vee x_8), \\ (f_7, f_8, f_9) &= (u_3 x_1, u_3 \bar{x}_1, u_3 x_2), \\ (f_{10}, f_{11}, f_{12}) &= (u_4, x_4, \bar{x}_4). \end{aligned}$$

Applying the four aggregation rules to these updates gives

$$\begin{aligned} P(f(x)) &= x_{10} + x_{11} + u_1 + x_{10} + x_{11} = u_1, \\ A(f(x)) &= (u_2 \vee x_7)(u_2 \vee \bar{x}_7)(u_2 \vee x_8) = u_2, \\ O(f(x)) &= (u_3 x_1) \vee (u_3 \bar{x}_1) \vee (u_3 x_2) = u_3, \\ M(f(x)) &= \text{maj}(u_4, x_4, \bar{x}_4) = u_4. \end{aligned}$$

The last equality holds because the last two arguments are opposite, leaving the first argument to determine the majority. Thus $hf = gh$. None of the twelve coordinate functions of f is constant.

All four target nodes interact. The wiring diagram of g and one lift of each of its ten edges are shown below. Each lift is an essential dependence obtained directly from the displayed coordinate functions of f .



Edge in $W(g)$	A lift in $W(f)$
$1 \rightarrow 1$	$1 \rightarrow 3$
$2 \rightarrow 1$	$4 \rightarrow 3$
$3 \rightarrow 1$	$7 \rightarrow 3$
$1 \rightarrow 2$	$1 \rightarrow 4$
$4 \rightarrow 2$	$10 \rightarrow 4$
$2 \rightarrow 3$	$4 \rightarrow 7$
$4 \rightarrow 3$	$10 \rightarrow 7$
$1 \rightarrow 4$	$1 \rightarrow 10$
$3 \rightarrow 4$	$7 \rightarrow 10$
$4 \rightarrow 4$	$10 \rightarrow 10$

The reverse implication fails. For instance, $10 \rightarrow 1$ is an edge of $W(f)$ because $f_1 = x_{10}$. Its endpoints have images 4 and 1, but $4 \rightarrow 1$ is absent from $W(g)$: the occurrences of x_{10} cancel in the first aggregate. Similarly, the source edges $7 \rightarrow 4$, $1 \rightarrow 7$, and $4 \rightarrow 11$ have images $3 \rightarrow 2$, $1 \rightarrow 3$, and $2 \rightarrow 4$, respectively, all absent from $W(g)$. The AND, OR, and majority computations above explain their disappearance. In particular, a compatible node map need not be a graph homomorphism between the wiring diagrams.

The construction is not limited to twelve nodes or to these four rules. Let $[n] = B_1 \sqcup \dots \sqcup B_m$ be any partition into nonempty blocks, and choose any nonconstant functions $H_a: \mathbb{F}_2^{B_a} \rightarrow \mathbb{F}_2$. Define h by equation (2.9). Independent choices in the blocks show that h is surjective, and its block assignment is compatible. For each a , choose states $s_a(0), s_a(1)$ with $H_a(s_a(t)) = t$. Given any target network g , define a source network by

$$f(x)|_{B_a} = s_a(g_a(h(x))), \quad a \in [m].$$

Then $h_a(f(x)) = g_a(h(x))$ for every a , so h is a contractive semiconjugacy. Both the number of blocks and their sizes can be arbitrarily large. The preceding example allows additional variation within the aggregate fibers, which accounts for its disappearing dependencies.

2.2.4 Lifting edges through node fibers

The examples suggest a one-sided relationship between wiring diagrams: every dependency in the factor should come from at least one dependency between the appropriate source blocks. Several source edges may give the same target edge, and some source dependencies may disappear under aggregation. For a node map ϕ , write

$$(\phi \times \phi)(E(W(f))) = \{(\phi(i), \phi(j)) : (i, j) \in E(W(f))\}.$$

We seek inclusion of $E(W(g))$ in this set. Equality, and hence preservation of every source edge, is not asserted.

Here *edge lifting* means the existence of an edge between the indicated node fibers. A graph fibration additionally requires a graph homomorphism and a unique incoming-edge lift at every prescribed vertex above its target [3, Definition 3.1].

Theorem 2.2.11 (Edge lifting). *Let $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ and $g: \mathbb{F}_2^m \rightarrow \mathbb{F}_2^m$ be Boolean networks. Suppose that $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is a contractive semiconjugacy admitting a compatible node map $\phi: [n] \rightarrow [m]$. Write $B_a = \phi^{-1}(a)$. Then*

$$a \longrightarrow b \text{ in } W(g) \implies \begin{array}{l} \text{there exist } i \in B_a \text{ and } j \in B_b \\ \text{such that } i \longrightarrow j \text{ in } W(f). \end{array} \quad (2.11)$$

Equivalently,

$$E(W(g)) \subseteq (\phi \times \phi)(E(W(f))).$$

Self-loops are included: a loop at a lifts to an edge whose two endpoints lie in B_a , and these endpoints need not be equal. Since compatibility already implies contraction, semiconjugacy and compatibility are sufficient hypotheses.

Proof. Suppose $a \rightarrow b$ belongs to $W(g)$. By essential dependence, there is $y \in \mathbb{F}_2^m$ such that

$$g_b(y) \neq g_b(y + e_a).$$

Surjectivity gives states $x, z \in \mathbb{F}_2^n$ with $h(x) = y$ and $h(z) = y + e_a$. Define x' by replacing only the coordinates of x in B_a :

$$x'_i = \begin{cases} z_i, & i \in B_a, \\ x_i, & i \notin B_a. \end{cases}$$

Proposition 2.2.9 shows that h_c depends only on B_c . Thus $h_a(x') = h_a(z) = y_a + 1$, while $h_c(x') = h_c(x) = y_c$ for every $c \neq a$. Consequently,

$$h(x') = y + e_a, \quad x'|_{[n] \setminus B_a} = x|_{[n] \setminus B_a}.$$

Applying the semiconjugacy identity gives

$$h_b(f(x)) = g_b(y) \neq g_b(y + e_a) = h_b(f(x')).$$

Since h_b depends only on B_b , the output states $f(x)$ and $f(x')$ must differ in at least one coordinate $j \in B_b$. Fix such a j .

Join x to x' by changing each coordinate in which they differ exactly once. All these coordinates belong to B_a . Write the resulting path as

$$x = x^{(0)}, x^{(1)}, \dots, x^{(\ell)} = x'.$$

Since $f_j(x) \neq f_j(x')$, there is a step t for which

$$f_j(x^{(t)}) \neq f_j(x^{(t+1)}).$$

This step has $x^{(t+1)} = x^{(t)} + e_i$ for some $i \in B_a$. It therefore witnesses an essential dependence of f_j on x_i , giving $i \rightarrow j$ in $W(f)$. By construction, $\phi(i) = a$ and $\phi(j) = b$, as required. The argument also applies when $a = b$. $\square$

Chapter 3

Calculus of Boolean Networks

3.1 Morphisms and Boolean observables

Throughout this chapter, $[n] = \{1, \dots, n\}$, and all sums and products of Boolean values are taken in $\mathbb{F}_2$. Dimensions are positive unless stated otherwise. We use the unsigned wiring diagrams of the preceding chapter, including self-loops.

Definition 3.1.1. Let $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ and $g: \mathbb{F}_2^m \rightarrow \mathbb{F}_2^m$ be Boolean networks. A *morphism* from f to g is a surjective Boolean map $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ satisfying

$$h \circ f = g \circ h$$

that is a contraction and admits a compatible node map in the sense of Definition 2.2.8. We write $h: f \rightarrow g$.

Morphisms are the state maps satisfying the hypotheses of Theorem 2.2.11. Compatibility already implies contraction. A compatible node map is a witness to this condition; it is not additional data in the morphism. In particular, two different compatible node maps for the same h do not define different morphisms.

When a witness $\phi: [n] \rightarrow [m]$ is chosen, we write

$$B_a = \phi^{-1}(a), \quad h_a(x) = H_a(x|_{B_a}), \quad a \in [m]. \quad (3.1)$$

By Proposition 2.2.9, the blocks are nonempty and the functions H_a are nonconstant. Conversely, any such block map is surjective and contractive. It is a morphism $f \rightarrow g$ exactly when it satisfies the displayed commutation identity.

Proposition 3.1.2. *Boolean networks and their morphisms form a category **BN**. Every morphism is an epimorphism. The isomorphisms are exactly the isometric conjugacies, and every endomorphism is an automorphism.*

Proof. The identity state map is a morphism, with the identity node map as a witness. Suppose $h: f \rightarrow g$ and $k: g \rightarrow r$ have compatible node maps $\phi: [n] \rightarrow [m]$ and $\psi: [m] \rightarrow [\ell]$. Changing input i either leaves h unchanged or changes only its coordinate $\phi(i)$. Applying k then either erases this change or changes only coordinate $\psi(\phi(i))$. Thus $\psi \circ \phi$ is compatible with $k \circ h$. The composite is surjective and

$$(k \circ h) \circ f = k \circ g \circ h = r \circ (k \circ h).$$

Associativity and the identity laws are those of functions. Surjectivity gives right cancellation, so every morphism is an epimorphism.

An invertible morphism is a bijection between cubes, so their dimensions agree. A surjective contraction between cubes of the same dimension is an isometry, as proved in Section 2.2. Conversely, the inverse of an isometric conjugacy is an isometric conjugacy and hence a morphism. The same argument applies to every endomorphism, since its source and target dimensions agree. $\square$

For the composite above, the block assigned to $c \in [\ell]$ is

$$(\psi \circ \phi)^{-1}(c) = \bigcup_{a \in \psi^{-1}(c)} B_a.$$

Consequently, successive reductions combine the blocks of the earlier reduction. A morphism can only decrease the number of nodes. A morphism between networks with the same number of nodes is an isometric conjugacy.

Example 3.1.3. Put $P = x_1 + x_2$ and $A = x_3x_4$, and define

$$\begin{aligned} h(x) &= (P, A), \\ f(x) &= (x_3, P + A + x_3, P, A), \\ g(y) &= (y_1 + y_2, y_1y_2). \end{aligned}$$

The blocks $B_1 = \{1, 2\}$ and $B_2 = \{3, 4\}$ give a compatible node map. Parity on the first block and AND on the second block are nonconstant, so h is surjective. Moreover,

$$h(f(x)) = (x_3 + P + A + x_3, PA) = (P + A, PA) = g(h(x)).$$

Thus $h: f \rightarrow g$ is a morphism. We will use it to compare the calculus before and after aggregation.

3.1.1 Pullback of observables

Definition 3.1.4. A *Boolean observable* on $\mathbb{F}_2^n$ is a function $q: \mathbb{F}_2^n \rightarrow \mathbb{F}_2$. Let $\mathcal{A}_n$ be the algebra of these functions, with pointwise addition and multiplication. For a Boolean map $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, its *pullback* is

$$h^*: \mathcal{A}_m \rightarrow \mathcal{A}_n, \quad h^*q = q \circ h.$$

Every observable satisfies $q^2 = q$. Negation and disjunction can be expressed in this algebra as $1 + q$ and $p \vee q = p + q + pq$. An observable q on the reduced network therefore gives an observable h^*q on the original network by evaluating it on the observed state.

Proposition 3.1.5. *If $h: f \rightarrow g$ is a morphism, then h^* is an injective unital algebra homomorphism. Its image consists exactly of the observables constant on every state fiber of h . Furthermore,*

$$f^*h^* = h^*g^*, \quad (k \circ h)^* = h^*k^*. \quad (3.2)$$

Proof. Composition with h preserves sums, products, and the constant function 1. If $q \circ h = r \circ h$, surjectivity implies $q = r$. A pullback is constant on fibers. Conversely, if p is constant on fibers, define $q(y)$ to be the common value of p on $h^{-1}(y)$. Then $p = h^*q$. The two identities follow from $hf = gh$ and associativity of composition. $\square$

Proposition 3.1.6. *Let h be a surjective compatible block map and let f be a Boolean network. The following conditions are equivalent:*

- (i) *There is a network g for which $h: f \rightarrow g$ is a morphism.*
- (ii) *The subalgebra $h^*(\mathcal{A}_m)$ is preserved by f^* .*

(iii) Each observable $h_a \circ f$ is constant on every fiber of h .

When these conditions hold, g is unique.

Proof. Condition (i) gives (ii) by equation (3.2). Since h_a is the pullback of the coordinate observable y_a , condition (ii) gives (iii). Under (iii), define $g_a(y) = h_a(f(x))$ for any $x \in h^{-1}(y)$. This is well defined and gives $gh = hf$. Surjectivity gives uniqueness. $\square$

This criterion expresses reduction in terms of measurements: the selected observables and every Boolean expression formed from them must be updatable using only their present values. It also gives a direct finite procedure. One checks compatibility of the selected coordinate functions and then checks that the aggregate update is constant on each state fiber.

3.2 Finite differences on the Boolean cube

3.2.1 Partial derivatives and the product rule

The finite differences used here belong to the classical Boolean calculus of differences; see [10]. The algebraic normal form and directional derivatives of Boolean functions are treated in [1, Section 2.2.1]. We give the identities needed below explicitly.

Definition 3.2.1. For $q \in \mathcal{A}_n$ and $i \in [n]$, the *Boolean partial derivative* of q in coordinate i is

$$D_i q(x) = q(x + e_i) + q(x). \quad (3.3)$$

The *essential support* of q is

$$\text{Ess}(q) = \{i \in [n] : D_i q \neq 0\}.$$

The derivative takes the value 1 precisely when changing the indicated coordinate changes the observable. In particular, $D_i q$ is independent of x_i . A coordinate outside $\text{Ess}(q)$ can be changed without affecting q , and changing such coordinates one at a time shows that q depends only on its essential support.

Proposition 3.2.2. For $p, q \in \mathcal{A}_n$ and $i, j \in [n]$,

$$D_i(p + q) = D_i p + D_i q, \quad (3.4)$$

$$D_i(pq) = pD_i q + qD_i p + (D_i p)(D_i q), \quad (3.5)$$

$$D_i^2 q = 0, \quad D_i D_j q = D_j D_i q. \quad (3.6)$$

Moreover, all partial derivatives of q vanish if and only if q is constant.

Proof. The sum rule follows immediately from the definition. Write

$$p(x + e_i) = p(x) + D_i p(x), \quad q(x + e_i) = q(x) + D_i q(x).$$

Multiplying and adding $p(x)q(x)$ gives the product rule. Applying D_i twice counts both endpoint values twice, giving zero. Applying $D_i D_j$ adds the values at the four vertices x , $x + e_i$, $x + e_j$, and $x + e_i + e_j$, independently of the order. Finally, vanishing derivatives make q constant along every cube edge, hence on the connected cube. $\square$

Example 3.2.3. For $q(x_1, x_2) = x_1 x_2$,

$$D_1 q = x_2, \quad D_2 q = x_1, \quad D_1 D_2 q = 1.$$

The correction term in equation (3.5) is necessary even for one variable. Since $x_1^2 = x_1$, its derivative is 1, whereas the two terms $x_1 D_1 x_1 + x_1 D_1 x_1$ cancel.

There is also an algebraic reason for this correction. If an $\mathbb{F}_2$ -linear operator $\delta: \mathcal{A}_n \rightarrow \mathcal{A}_n$ satisfied the ordinary Leibniz rule, then

$$\delta(q) = \delta(q^2) = q\delta(q) + q\delta(q) = 0$$

for every q . Thus ordinary derivations of this algebra are zero. Boolean partial derivatives record finite changes and satisfy equation (3.5) instead.

3.2.2 Higher derivatives and exact expansion

For $I \subseteq [n]$, define $D_I = \prod_{i \in I} D_i$, with $D_\emptyset$ the identity operator. Commutativity makes the order irrelevant. Repeated use of the definition gives

$$D_I q(x) = \sum_{T \subseteq I} q\left(x + \sum_{i \in T} e_i\right). \quad (3.7)$$

Thus $D_I q$ is independent of the coordinates indexed by I . A nonzero higher derivative records an interaction involving all of those coordinates.

Theorem 3.2.4 (Boolean Taylor expansion). *For $q \in \mathcal{A}_n$ and $x, v \in \mathbb{F}_2^n$,*

$$q(x + v) = \sum_{I \subseteq [n]} D_I q(x) \prod_{i \in I} v_i. \quad (3.8)$$

Empty products are 1. Consequently, every Boolean observable has a unique multilinear expression

$$q(x) = \sum_{I \subseteq [n]} c_I \prod_{i \in I} x_i, \quad c_I = D_I q(\mathbf{0}). \quad (3.9)$$

Proof. Translation by e_i is the operator $\text{Id} + D_i$. Translation by v is therefore

$$\prod_{\{i: v_i=1\}} (\text{Id} + D_i).$$

Expanding this product and applying it to q proves equation (3.8). Taking the base point to be $\mathbf{0}$ proves existence in equation (3.9). Differentiating a multilinear monomial in the coordinates of I and evaluating at $\mathbf{0}$ gives 1 exactly when that monomial has support I . This determines every coefficient and proves uniqueness. $\square$

Expression (3.9) is the *algebraic normal form*. For nonzero q , its algebraic degree is the largest $|I|$ with $c_I = 1$; nonzero constants have degree 0. We set $\deg 0 = -\infty$. Equivalently, a nonconstant q has degree equal to the largest $|I|$ for which $D_I q \neq 0$.

Example 3.2.5. The second update in Example 3.1.3 is

$$f_2 = x_1 + x_2 + x_3 x_4 + x_3.$$

Its first derivatives are $1, 1, x_4 + 1, x_3$, and its only nonzero derivative of order two is $D_{\{3,4\}} f_2 = 1$. Thus its wiring diagram records four essential inputs, while the higher derivative singles out the interaction between inputs 3 and 4.

3.2.3 Integration and reconstruction

Proposition 3.2.6 (Integration on cube edges). *Given functions $p_1, \dots, p_n \in \mathcal{A}_n$, there is an observable q satisfying $D_i q = p_i$ for every i if and only if*

$$D_i p_i = 0, \quad D_i p_j = D_j p_i \quad \text{for all } i, j. \quad (3.10)$$

Such a q is uniquely determined by its value at $\mathbf{0}$.

Proof. Necessity follows from equation (3.6). For sufficiency, assign the value $p_i(x)$ to the edge joining x and $x + e_i$. The condition $D_i p_i = 0$ makes this independent of the endpoint used. The other condition says that the sum around every coordinate square is zero:

$$p_i(x) + p_j(x + e_i) + p_i(x + e_j) + p_j(x) = 0.$$

Fix $q(\mathbf{0}) = c$. For any x , take a path from $\mathbf{0}$ to x that changes each coordinate with $x_i = 1$ once, and define $q(x)$ as c plus the sum of its edge values. Two orders of these coordinate changes differ by adjacent interchanges. Each interchange crosses a coordinate square, so the sum is independent of the order.

If $x_i = 0$, a path to x followed by its i th edge is an allowed path to $x + e_i$. Hence $q(x + e_i) + q(x) = p_i(x)$. If $x_i = 1$, apply the same argument to $x + e_i$ and use endpoint independence. Uniqueness follows because a function with all derivatives zero is constant. $\square$

For example, along a coordinate path $x = x^{(0)}, \dots, x^{(r)} = z$, with $x^{(t+1)} = x^{(t)} + e_{i_t}$, the fundamental telescoping identity is

$$q(z) + q(x) = \sum_{t=0}^{r-1} D_{i_t} q(x^{(t)}). \quad (3.11)$$

Consequently, all partial derivatives of a network, together with one output value, determine the entire network. The wiring diagram alone retains only which derivatives are nonzero somewhere, so it contains less information.

3.2.4 Jacobians and local wiring diagrams

Definition 3.2.7. For a Boolean map $F: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$, its *Boolean Jacobian* at x is the $r \times n$ matrix

$$J_F(x)_{ji} = D_i F_j(x).$$

For a network f , the *local wiring diagram* $W_x(f)$ has an edge $i \rightarrow j$ exactly when $J_f(x)_{ji} = 1$.

The global wiring diagram is the union of the local diagrams:

$$E(W(f)) = \bigcup_{x \in \mathbb{F}_2^n} E(W_x(f)).$$

This is a union of edge sets, rather than a sum of matrices over $\mathbb{F}_2$. Matrix addition can cancel two occurrences of the same edge.

For a compatible block map h , each column of $J_h(x)$ is either zero or the basis vector $e_{\phi(i)}$. Let

$$S_a = \text{Ess}(h_a) \subseteq B_a. \quad (3.12)$$

The sets S_a are nonempty and pairwise disjoint. Unlike the blocks B_a , they are determined by h itself. Coordinates outside their union are invisible to every output of h .

3.3 Differentiation through a morphism

3.3.1 Composition and its correction terms

Proposition 3.3.1 (Exact composition rule). *For $F: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$ and $q \in \mathcal{A}_r$,*

$$D_i(q \circ F)(x) = \sum_{\emptyset \neq T \subseteq [r]} (D_T q)(F(x)) \prod_{j \in T} D_i F_j(x). \quad (3.13)$$

If F is a contraction, then for every Boolean map Q of the appropriate dimension,

$$J_{Q \circ F}(x) = J_Q(F(x)) J_F(x), \quad (3.14)$$

with matrix multiplication over $\mathbb{F}_2$.

Proof. Apply Theorem 3.2.4 to q at $F(x)$ with displacement $F(x + e_i) + F(x)$. Its j th coordinate is $D_i F_j(x)$. Adding $q(F(x))$ cancels the term indexed by $\emptyset$ and proves equation (3.13). If F is a contraction, this displacement has at most one nonzero coordinate. Every product indexed by $|T| \geq 2$ vanishes, leaving the matrix identity. $\square$

In particular, if $h: f \rightarrow g$ is a morphism, the composite $g \circ h$ satisfies the ordinary matrix formula. The other composite $h \circ f$ can require higher derivatives, since f need not be a contraction.

Corollary 3.3.2. *Let $h: f \rightarrow g$ have compatible node map ϕ , and let $a = \phi(i)$. For every target coordinate b ,*

$$D_i h_a(x) D_a g_b(h(x)) = \sum_{\emptyset \neq T \subseteq S_b} (D_T h_b)(f(x)) \prod_{j \in T} D_i f_j(x). \quad (3.15)$$

Equivalently,

$$J_g(h(x)) J_h(x) = J_h(f(x)) J_f(x) + R(x), \quad (3.16)$$

where

$$R(x)_{bi} = \sum_{\substack{T \subseteq S_b \\ |T| \geq 2}} (D_T h_b)(f(x)) \prod_{j \in T} D_i f_j(x).$$

The correction R vanishes if every aggregation function is affine, or if f is a contraction.

Proof. Differentiate $g_b \circ h = h_b \circ f$ using Proposition 3.3.1. Only coordinate a can change through h when input i changes. Derivatives of h_b involving a coordinate outside S_b vanish. Separating singleton sets from larger sets gives equation (3.16). An affine function has no derivatives of order at least two, and a contraction has at most one nonzero entry in each Jacobian column. $\square$

Example 3.3.3. Put $p = x_1 x_2$ and consider

$$h(x) = p, \quad f(x) = (p, p), \quad g(y) = y.$$

Since $p^2 = p$, we have $hf = gh$. The single block $\{1, 2\}$ makes h a morphism. At $x = (0, 1)$,

$$J_h(x) = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad J_f(x) = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad J_h(f(x)) = \begin{pmatrix} 0 & 0 \end{pmatrix}.$$

Hence $J_g(h(x)) J_h(x) = \begin{pmatrix} 1 & 0 \end{pmatrix}$, while $J_h(f(x)) J_f(x) = 0$. The term involving $D_{\{1,2\}} h = 1$ supplies the missing row $\begin{pmatrix} 1 & 0 \end{pmatrix}$.

3.3.2 An exact chain rule for all partial derivatives

Theorem 3.3.4 (Block chain rule). *Let h be a surjective compatible block map. For a nonempty set $I \subseteq [n]$, put*

$$A = \phi(I), \quad I_a = I \cap B_a \quad (a \in A).$$

Then, for every $q \in \mathcal{A}_m$,

$$D_I(h^*q) = h^*(D_Aq) \prod_{a \in A} D_{I_a} h_a. \quad (3.17)$$

Moreover,

$$D_I(h^*q) \neq 0 \iff \begin{cases} D_Aq \neq 0, \\ D_{I_a} h_a \neq 0 \text{ for every } a \in A. \end{cases} \quad (3.18)$$

Proof. Expand q as a multilinear polynomial in the active coordinates y_a , $a \in A$, with coefficient functions in the remaining coordinates:

$$q(y) = \sum_{T \subseteq A} q_T(y|_{[m] \setminus A}) \prod_{a \in T} y_a.$$

The coefficient q_A is D_Aq . After substitution by h , a term missing an active coordinate a is independent of the entire block B_a . Since I_a is nonempty, applying D_I kills that term. In the remaining term, the coefficient is independent of all active blocks, and the other factors depend on disjoint blocks. Differentiating each factor in its own block gives equation (3.17).

If any factor on its right is identically zero, the result is zero. Conversely, suppose all indicated derivatives are nonzero. Choose the active blocks independently so that each $D_{I_a} h_a$ is 1. The function D_Aq is independent of y_a for $a \in A$. Choose its remaining arguments to give value 1, and realize those arguments using surjectivity of the remaining block functions. These independent choices give an input where the product is 1. $\square$

Corollary 3.3.5. *For every observable q ,*

$$\text{Ess}(h^*q) = \bigcup_{a \in \text{Ess}(q)} S_a. \quad (3.19)$$

*In particular, every essential input of h_a remains essential in h^*q whenever q depends essentially on y_a .*

Proof. Take $I = \{i\}$ in equation (3.18). $\square$

The formula covers differentiation in several inputs of the same block. For instance, if $i, j \in B_a$ are distinct, then

$$D_i D_j(h^*q) = h^*(D_aq) D_i D_j h_a.$$

Thus repeated differentiation within one block measures the aggregation rule, while differentiation across distinct blocks also measures interactions in the target observable.

Example 3.3.6. In Example 3.1.3, let $q(y) = y_1 y_2$. Then

$$h^*q = (x_1 + x_2)x_3x_4.$$

Since $D_{\{1,2\}}q = 1$,

$$D_{\{1,3\}}(h^*q) = x_4, \quad D_{\{1,3,4\}}(h^*q) = 1.$$

The third derivative combines the interaction between the two target coordinates with the second derivative of the AND aggregation on B_2 .

Proposition 3.3.7 (Degree under pullback). *Write $d_a = \deg h_a \geq 1$ and let the nonzero observable q have algebraic normal form $q(y) = \sum_A c_A \prod_{a \in A} y_a$. Then*

$$\deg(h^*q) = \max_{\{A: c_A=1\}} \sum_{a \in A} d_a. \quad (3.20)$$

*In particular, $\deg(h^*q) \geq \deg q$.*

Proof. Substituting the polynomials h_a gives the upper bound. Choose A of greatest indicated weight and choose a monomial of degree d_a in each h_a for $a \in A$. Since the blocks are disjoint, their product has degree $\sum_{a \in A} d_a$ and uses variables in every block indexed by A .

For another target monomial to contribute this same source monomial, its support must contain A . A strict superset with nonzero coefficient would have greater weight, because every d_a is positive. Within the term indexed by A , disjointness of the blocks makes the chosen product unique. It therefore cannot cancel, proving the lower bound. $\square$

This statement concerns pullback of observables. Nonlinear aggregation of the outputs of f can still produce target updates of higher degree than any individual source update.

Example 3.3.8. Define

$$h(x) = (x_1, x_2, x_3 x_4), \quad f(x) = (x_1, x_2, x_1, x_2),$$

and $g(y) = (y_1, y_2, y_1 y_2)$. The blocks $\{1\}$, $\{2\}$, and $\{3, 4\}$ make h a morphism, since

$$h(f(x)) = (x_1, x_2, x_1 x_2) = g(h(x)).$$

Every coordinate function of f is linear, whereas $D_{\{1,2\}} g_3 = 1$. Neither f_3 nor f_4 has this mixed interaction. It appears when their outputs are multiplied by h_3 .

3.3.3 Sections and recovery of the reduced calculus

Proposition 3.3.9 (Isometric sections). *Every surjective compatible block map $h: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ admits a section $s: \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$ and indices $i_a \in S_a$ such that*

$$h \circ s = \text{Id}, \quad s(y + e_a) = s(y) + e_{i_a}. \quad (3.21)$$

The section preserves Hamming distances. One may prescribe any essential coordinate $i_a \in S_a$ in each block.

Proof. For each prescribed $i_a \in S_a$, essential dependence provides adjacent block states u_a^0, u_a^1 differing only in coordinate i_a and satisfying $H_a(u_a^t) = t$. Define

$$s(y)|_{B_a} = u_a^{y_a}.$$

Independent blocks give $h(s(y)) = y$. Changing y_a changes exactly coordinate i_a of $s(y)$, which proves the second identity and distance preservation. $\square$

The section selects a coordinate cube inside the source state space. It need not preserve the dynamics, and if $m < n$ it is not surjective, so it is not a morphism in **BN**.

Corollary 3.3.10. *Let s and i_a be as above, and put $I_A = \{i_a : a \in A\}$. For every $A \subseteq [m]$,*

$$(D_A q)(y) = D_{I_A}(h^*q)(s(y)). \quad (3.22)$$

If $h: f \rightarrow g$ is a morphism, then

$$g = h \circ f \circ s, \quad D_A g_b(y) = D_{I_A}(h_b \circ f)(s(y)). \quad (3.23)$$

Proof. For every a , $D_{i_a} h_a(s(y)) = 1$. The first identity follows from Theorem 3.3.4, including the empty set case by $hs = \text{Id}$. The remaining identities follow by applying it to $q = g_b$ and using $g_b h = h_b f$. $\square$

For example, all algebraic normal form coefficients of g_b can be read as $D_{I_A}(h_b f)(s(\mathbf{0}))$. The values of f on the selected coordinate cube therefore suffice to reconstruct g and all of its derivatives.

3.3.4 Influence and the distribution of observations

Definition 3.3.11. For an observable q on $\mathbb{F}_2^n$, the *influence* of coordinate i is

$$\text{Inf}_i(q) = 2^{-n} |\{x : D_i q(x) = 1\}|.$$

A Boolean function is *balanced* if it takes each value on exactly half of its inputs.

Influence measures the fraction of input states at which an edge is active. Its value is positive exactly when the corresponding coordinate is essential. Let X be uniformly distributed on $\mathbb{F}_2^n$, and write $Y = h(X)$ and μ_h for the distribution of Y . Disjointness of the blocks makes the variables Y_a independent, but they need not be uniformly distributed.

Proposition 3.3.12. For $i \in B_a$,

$$\text{Inf}_i(h^* q) = \text{Inf}_i(h_a) \Pr_{Y \sim \mu_h} [D_a q(Y) = 1]. \quad (3.24)$$

If every H_a is balanced, then Y is uniform and the last factor is $\text{Inf}_a(q)$.

Proof. The first derivative chain rule expresses $D_i(h^* q)$ as the product of $D_i h_a$ and $(D_a q) \circ h$. The first factor depends only on B_a . Since $D_a q$ is independent of y_a , the second factor depends only on the other blocks. The two events are independent under the uniform distribution on source states, giving the formula. Balanced independent outputs give the uniform distribution on $\mathbb{F}_2^m$. $\square$

Example 3.3.13. For $h = (x_1 + x_2, x_3 x_4)$ and $q(y) = y_1 y_2$, the observed second coordinate equals 1 with probability 1/4. Hence

$$\text{Inf}_1(h^* q) = \frac{1}{4}, \quad \text{Inf}_1(h_1) = 1, \quad \text{Inf}_1(q) = \frac{1}{2}.$$

Uniform averaging on the target would give the wrong pullback influence. Equation (3.24) uses the distribution actually induced by the aggregation.

3.4 Wiring diagrams, paths, and feedback

3.4.1 Lifting from every essential input

The exact support formula strengthens Theorem 2.2.11. It identifies which source nodes can initiate a lift and permits successive edges to be lifted consistently.

Theorem 3.4.1 (Essential edge lifting). *Let $h: f \rightarrow g$ be a morphism and let $S_a = \text{Ess}(h_a)$. If $a \rightarrow b$ is an edge of $W(g)$, then*

$$\text{for every } i \in S_a \text{ there is } j \in S_b \text{ with } i \rightarrow j \text{ in } W(f). \quad (3.25)$$

There is also a simultaneous local version. For a section s and indices i_a from Proposition 3.3.9,

$$a \rightarrow b \text{ in } W_y(g) \implies i_a \rightarrow j \text{ in } W_{s(y)}(f) \text{ for some } j \in S_b. \quad (3.26)$$

Proof. If $a \rightarrow b$ in $W(g)$, Corollary 3.3.5 gives

$$S_a \subseteq \text{Ess}(g_b \circ h) = \text{Ess}(h_b \circ f).$$

The function h_b depends only on the coordinates in S_b . Therefore,

$$\text{Ess}(h_b \circ f) \subseteq \bigcup_{j \in S_b} \text{Ess}(f_j).$$

Every $i \in S_a$ must belong to at least one support on the right, which proves equation (3.25).

For the local statement, put $x = s(y)$. Since $h(x + e_{i_a}) = y + e_a$, an edge $a \rightarrow b$ in $W_y(g)$ gives

$$h_b(f(x + e_{i_a})) + h_b(f(x)) = g_b(y + e_a) + g_b(y) = 1.$$

Hence $f(x + e_{i_a})$ and $f(x)$ differ in at least one coordinate $j \in S_b$. This gives the required edge at x . The same section works for all edges and all target states. $\square$

Example 3.4.2. The local statement requires suitable representatives of the state fibers. In Example 3.3.3, at $x = (0, 0)$ both rows of $J_f(x)$ are zero, while $g(y) = y$ has a local self-loop at $h(x) = 0$. Thus local lifting at every source point above y would be false. The section $s(y) = (y, 1)$ chooses representatives at which the target loop does lift.

Nor can the endpoint of the lifted edge be prescribed arbitrarily. For

$$h(x_1, x_2) = x_1 + x_2, \quad f(x_1, x_2) = (x_1 + x_2, 0), \quad g(y) = y,$$

both source nodes are essential in h , and both have an edge to node 1. No edge enters node 2. The theorem specifies a starting node and guarantees some suitable endpoint.

Corollary 3.4.3. Let $d_f^+(i)$ and $d_g^+(a)$ denote out-degrees in the global wiring diagrams, counting a self-loop once. Then

$$d_f^+(i) \geq d_g^+(a) \quad (i \in S_a), \quad |E(W(f))| \geq \sum_{a=1}^m |S_a| d_g^+(a). \quad (3.27)$$

Proof. Distinct out-neighbors b of a require endpoints in disjoint sets S_b , so their lifts from i are distinct. Sum this bound over all essential source nodes. Edges from source nodes ignored by h can only increase the total. $\square$

3.4.2 Paths, cycles, and strongly connected components

Theorem 3.4.4. Let $h: f \rightarrow g$ be a morphism.

- (i) Every directed walk $a_0 \rightarrow \cdots \rightarrow a_r$ in $W(g)$ lifts, from any prescribed $i_0 \in S_{a_0}$, to a directed walk $i_0 \rightarrow \cdots \rightarrow i_r$ in $W(f)$ with $i_t \in S_{a_t}$. If the target vertices are distinct, the lifted vertices are distinct.
- (ii) Every simple directed cycle of length r in $W(g)$ lifts to a simple directed cycle of length kr in $W(f)$ that goes around the target cycle k times, where

$$1 \leq k \leq \min_{a \text{ on the target cycle}} |S_a|.$$

(iii) For every strongly connected component C of $W(g)$, there is a strongly connected subgraph of $W(f)$ contained in $\bigcup_{a \in C} S_a$ and meeting every S_a , $a \in C$.

Proof. For (i), repeatedly apply Theorem 3.4.1, using the endpoint of one lift as the starting point of the next. Disjointness of the supports gives the assertion about distinct vertices.

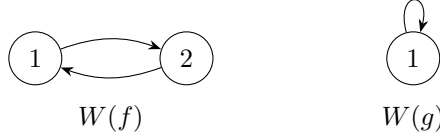
For (ii), retain the vertices above the target cycle and only source edges that follow its successive block labels. Every retained vertex has an outgoing edge in this finite graph. Following outgoing edges until a vertex repeats produces a simple directed cycle. Its labels advance around the target cycle, so its length is kr for some positive integer k . It visits each support on that cycle exactly k times, proving the bound. This also applies to a target self-loop, for which $r = 1$.

For (iii), retain vertices in $\bigcup_{a \in C} S_a$ and source edges whose labels form an edge of the target subgraph on C . Choose a strongly connected component K of this finite graph with no outgoing edge to another component. Such a component exists by following edges between components until none remain. If K meets S_a and $a \rightarrow b$ is an edge inside C , essential edge lifting provides an edge from a vertex of K to S_b . This edge cannot leave K , so K also meets S_b . The labels represented in K are therefore a nonempty subset of C closed under following edges. Strong connectivity of C forces this subset to be all of C . $\square$

Example 3.4.5. Let

$$f(x_1, x_2) = (x_2, x_1), \quad h(x_1, x_2) = x_1 + x_2, \quad g(y) = y.$$

This is a morphism. The target wiring diagram has a self-loop, while the source has a directed cycle of length two and no self-loop:



Thus lifting a wiring cycle preserves the cyclic order of block labels, but can increase its length.

Corollary 3.4.6. A morphism has the following consequences for wiring diagrams:

- (i) If $W(f)$ is acyclic, then $W(g)$ is acyclic.
- (ii) Any family of vertex-disjoint directed cycles in $W(g)$ lifts to a family of vertex-disjoint directed cycles in $W(f)$.
- (iii) The minimum number of vertices whose removal makes $W(g)$ acyclic is at most the corresponding number for $W(f)$.

Proof. The first assertion follows from cycle lifting. Disjoint target cycles use disjoint essential supports, giving the second. For the third, let K be a set of source vertices meeting every directed source cycle. Put

$$L = \{a \in [m] : S_a \cap K \neq \emptyset\}.$$

A target cycle avoiding L would lift to a source cycle avoiding K , which is impossible. Thus L meets every target cycle. The supports are disjoint, so $|L| \leq |K|$. $\square$

All these conclusions concern the essential supports of the observation. They do not imply that every source edge has an image in $W(g)$, or that the compatible node map is a graph homomorphism. The disappearing dependencies in Example 2.2.10 remain possible.

3.5 Perturbations and iteration

3.5.1 Exact propagation of a finite perturbation

Definition 3.5.1. For a Boolean map $F: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$, define

$$\Delta F(x; v) = F(x + v) + F(x), \quad TF(x, v) = (F(x), \Delta F(x; v)).$$

The first component records the updated reference state, and the second records the difference between the two updated states.

The Jacobian gives $\Delta F(x; e_i) = J_F(x)e_i$. For a general displacement, Theorem 3.2.4 gives

$$\Delta F(x; v) = \sum_{\emptyset \neq I \subseteq [n]} D_I F(x) \prod_{i \in I} v_i, \quad (3.28)$$

where differentiation of a vector map is componentwise. Thus $J_F(x)v$ need not describe a simultaneous change in several coordinates.

Proposition 3.5.2. For composable Boolean maps F, Q ,

$$\Delta(Q \circ F)(x; v) = \Delta Q(F(x); \Delta F(x; v)), \quad T(Q \circ F) = TQ \circ TF. \quad (3.29)$$

In particular, a morphism $h: f \rightarrow g$ satisfies

$$Th \circ Tf = Tg \circ Th. \quad (3.30)$$

Proof. Since $F(x) + \Delta F(x; v) = F(x + v)$, the right side of the first identity is $Q(F(x + v)) + Q(F(x))$. The identity for T includes this equality and the ordinary composition of reference states. Applying it to $hf = gh$ gives the last identity. $\square$

The maps TF give an exact calculus on pairs of states written as a reference state and a displacement. Equation (3.30) is an identity of Boolean maps. It does not assert that Th is a morphism under Definition 3.1.1: the change of coordinates from state pairs to reference states and displacements need not preserve Hamming distances.

Corollary 3.5.3. Write

$$K_h(x) = \{v \in \mathbb{F}_2^n : h(x + v) = h(x)\}.$$

If $h: f \rightarrow g$ is a morphism, then

$$v \in K_h(x) \implies \Delta f(x; v) \in K_h(f(x)).$$

In particular, an initially invisible difference remains invisible after every number of updates.

Proof. If $h(x + v) = h(x)$, semiconjugacy gives $h(f(x + v)) = h(f(x))$. This is the asserted condition on $\Delta f(x; v)$. Iterate the same argument. $\square$

Example 3.5.4. For $h(x_1, x_2) = x_1 x_2$, the set $K_h(0, 0)$ is $\{00, 10, 01\}$. It is not a vector subspace: $10 + 01 = 11$ is visible. Although $J_h(0, 0) = 0$, changing both coordinates changes h . The second derivative detects this effect. In Example 3.3.3, the full set of invisible differences is preserved by the dynamics as described in the corollary.

3.5.2 Bounds on amplification

For a Boolean map F define its Hamming Lipschitz constant by

$$L(F) = \max_{x \neq z} \frac{d_H(F(x), F(z))}{d_H(x, z)}.$$

Following shortest cube paths shows that this maximum can be computed on edges:

$$L(F) = \max_{x, i} d_H(F(x + e_i), F(x)) = \max_{x, i} \sum_j D_i F_j(x). \quad (3.31)$$

The last sum is taken in the integers. Indeed, the right side bounds the change at every step of a shortest path, giving the upper bound for all pairs; each edge itself is one of those pairs.

Proposition 3.5.5. *For a morphism $h: f \rightarrow g$ and every $t \geq 0$,*

$$L(g^t) \leq L(f^t). \quad (3.32)$$

Consequently, a contractive source update has a contractive target update, and an isometric source update has an isometric target update.

Proof. Let s be an isometric section of h . Iterating semiconjugacy gives $g^t = h f^t s$. For distinct y, z ,

$$\begin{aligned} d_H(g^t(y), g^t(z)) &\leq d_H(f^t(s(y)), f^t(s(z))) \\ &\leq L(f^t) d_H(s(y), s(z)) \\ &= L(f^t) d_H(y, z). \end{aligned}$$

If f is isometric, it is surjective. Since $gh = hf$ and both f and h are surjective, g is surjective as well. The bound makes g a contraction; a surjective contraction of a cube to itself is an isometry. $\square$

3.5.3 Periodic orbits and basins

For a finite network, every forward trajectory eventually enters a periodic orbit. Its *transient length* is the least number of updates needed to reach a periodic state. The *basin* of a periodic orbit is the set of states whose trajectories eventually enter it.

Theorem 3.5.6. *Let $h: f \rightarrow g$ be a morphism, and put $N_y = |h^{-1}(y)|$.*

- (i) *A source periodic orbit of length p maps onto a target periodic orbit of length q dividing p . Every target periodic orbit has at least one periodic source orbit above it.*
- (ii) *If a source periodic orbit lies above a target orbit D of length q , its length is kq for an integer satisfying*

$$1 \leq k \leq \min_{y \in D} N_y.$$

- (iii) *The transient length of $h(x)$ is at most the transient length of x .*

- (iv) *The inverse image of the basin of D is the union of the basins of all source periodic orbits whose images are D .*

Proof. If $f^p(x) = x$, then $g^p(h(x)) = h(x)$, so the target period divides p . For existence in the other direction, take a target point $y \in D$ of period q . The map f^q sends the nonempty finite set $h^{-1}(y)$ to itself and therefore has a periodic point there. This point is periodic under f and its orbit maps onto D .

If its f -period is $p = kq$, the successive source points have target labels cycling through D . Exactly k distinct points of the source orbit lie over each $y \in D$. This proves (ii).

Once a source trajectory reaches a periodic state, its image is periodic, proving (iii). Every source trajectory eventually reaches a unique source periodic orbit, whose image is the eventual target periodic orbit. This gives (iv). $\square$

Example 3.5.7. The projection $h(y, z) = y$ is a morphism from

$$f(y, z) = (y, z + 1) \quad \text{to} \quad g(y) = y.$$

Both target states are fixed, but the source has no fixed point. Each target fixed point lifts to a source cycle of length two. In particular, surjectivity on periodic states does not imply surjectivity on fixed points.

For all $t \geq 0$, surjectivity also gives

$$h(f^t(\mathbb{F}_2^n)) = g^t(\mathbb{F}_2^m).$$

Every identity between source iterates descends: if $f^r = f^s$, then $g^r h = h f^r = h f^s = g^s h$, and hence $g^r = g^s$. In particular, if an iterate of f is constant, the same iterate of g is constant. These are consequences for the phase dynamics, distinct from the wiring cycles considered in Section 3.4.

3.6 Scalar factors and dynamical observables

3.6.1 First integrals

Definition 3.6.1. A Boolean observable q is a *first integral* of f if $q \circ f = q$. Write

$$\mathcal{I}(f) = \{q \in \mathcal{A}_n : q \circ f = q\}.$$

The values of a first integral are constant along trajectories. The set $\mathcal{I}(f)$ is a subalgebra of $\mathcal{A}_n$, since updating preserves sums and products of invariant observables.

Proposition 3.6.2. *If f has c periodic orbits, then its first integrals are exactly the observables constant on the basin of each periodic orbit. Consequently, $\mathcal{I}(f)$ has 2^c elements and has dimension c as an $\mathbb{F}_2$ -vector space. A morphism $h: f \rightarrow g$ restricts to an injective algebra map*

$$h^*: \mathcal{I}(g) \rightarrow \mathcal{I}(f).$$

Proof. The identity $q(f(x)) = q(x)$ forces one value on an entire periodic orbit and the same value on every trajectory entering it. Conversely, assigning one Boolean value to each basin gives an invariant observable. The characteristic functions of the c basins form a basis, giving the cardinality and dimension. Finally, if $qg = q$, then $(qh)f = qgh = qh$, and injectivity follows from Proposition 3.1.5. $\square$

Thus the number of target periodic orbits is at most the number of source periodic orbits. Pullback identifies which unions of source basins are distinguishable by invariant observations on the target.

3.6.2 Classification of one-node factors

A nonconstant observable $u: \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is automatically surjective, contractive, and compatible with the unique node map $[n] \rightarrow [1]$. Hence it is a morphism to a one-node network g precisely when $uf = gu$. There are four possible update functions $g: \mathbb{F}_2 \rightarrow \mathbb{F}_2$.

Theorem 3.6.3. *Let f have c periodic orbits, and put $N = 2^n$ and $M = |f(\mathbb{F}_2^n)|$.*

- (i) *Morphisms to $g(t) = t$ are the nonconstant first integrals. There are $2^c - 2$ of them, so such a factor exists exactly when $c \geq 2$.*
- (ii) *A morphism to $g(t) = t + 1$ exists exactly when every periodic orbit of f has even length. In that case there are 2^c such morphisms.*
- (iii) *There are $2^{N-M} - 1$ morphisms to $g(t) = 0$, and the same number to $g(t) = 1$. Such a factor exists exactly when f is not bijective.*

Proof. Part (i) follows from Proposition 3.6.2 after excluding the two constant observables.

For (ii), the equation is $u(f(x)) = u(x) + 1$. Along a periodic orbit the values must alternate. Returning to the initial state is consistent exactly when the orbit length is even. In that case choose either value at one point of each periodic orbit and extend by alternation around the orbit. For a state whose trajectory first enters this orbit after r steps at z , define $u(x) = u(z) + (r \bmod 2)$. This gives the required identity at every state. There are two choices per basin and no additional choices. Every such observable is nonconstant because each periodic orbit contains both values.

For the constant-zero target, the equation $u(f(x)) = 0$ says exactly that u vanishes on the image of f . Its values at the $N - M$ remaining states are arbitrary, except that at least one must be 1 for surjectivity. This gives $2^{N-M} - 1$ choices. Complementing u gives the corresponding count for the constant-one target. A self-map of a finite set fails to be bijective exactly when its image omits a state. $\square$

Corollary 3.6.4. *Every Boolean network with at least one node has a one-node factor in **BN**.*

Proof. If f is not bijective, use a constant target. If it is bijective and has at least two periodic orbits, use the identity target. Otherwise its phase space is a single cycle of length 2^n , which is even because $n \geq 1$. The complementation target then gives a factor. $\square$

Example 3.6.5. For $f(y, z) = (y, z + 1)$, the observable $u(y, z) = y$ gives the identity factor from Example 3.5.7. The observable $v(y, z) = z$ gives the complementation factor. The first records which periodic orbit contains the state, and the second alternates along that orbit. Both observations are morphisms, but they retain different dynamical information.

3.7 Reconstruction and the information in a morphism

3.7.1 Recovering the state map from observables

The pullback of a morphism determines the morphism itself: its coordinate functions are the pullbacks of the target coordinate observables. The converse gives an algebraic description of exactly which observable embeddings come from morphisms.

Theorem 3.7.1. *Let $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ and $g: \mathbb{F}_2^m \rightarrow \mathbb{F}_2^m$. Morphisms $h: f \rightarrow g$ correspond bijectively to injective unital $\mathbb{F}_2$ -algebra homomorphisms*

$$\alpha: \mathcal{A}_m \rightarrow \mathcal{A}_n$$

satisfying

- (i) $f^* \alpha = \alpha g^*$;
- (ii) *the essential supports of $\alpha(y_1), \dots, \alpha(y_m)$ are pairwise disjoint.*

The correspondence is $\alpha = h^$.*

Proof. A morphism gives these properties by Proposition 3.1.5 and compatibility. Conversely, put $h_a = \alpha(y_a)$ and $h = (h_1, \dots, h_m)$. Every observable is a polynomial in the coordinate observables by Theorem 3.2.4. Since α preserves constants, sums, and products, it follows that $\alpha(q) = q \circ h$ for every q .

If h omitted a state $z \in \mathbb{F}_2^m$, the nonzero observable

$$\delta_z(y) = \prod_{a=1}^m (1 + y_a + z_a)$$

would vanish after composition with h , contradicting injectivity of α . Thus h is surjective, and in particular each h_a is nonconstant. Assign every coordinate in $\text{Ess}(h_a)$ to node a , and assign any unused coordinates arbitrarily. Pairwise disjointness makes this a well-defined compatible node map; nonconstancy makes it surjective. Finally, applying condition (i) to y_a gives $h_a \circ f = g_a \circ h$ for every a . Hence h is a morphism, uniquely determined by α . $\square$

Together with equation (3.2), this theorem describes composition of reductions entirely through embeddings of observable algebras, with the direction of the arrows reversed.

3.7.2 All source dynamics above a fixed reduction

Theorem 3.7.2 (Freedom inside state fibers). *Fix a surjective compatible block map h and a target network g . Write*

$$F_y = h^{-1}(y), \quad N_y = |F_y|.$$

The source networks f for which $h: f \rightarrow g$ is a morphism are exactly the choices of functions

$$f|_{F_y}: F_y \rightarrow F_{g(y)}, \quad y \in \mathbb{F}_2^m. \quad (3.33)$$

These choices are independent, so their total number is

$$\prod_{y \in \mathbb{F}_2^m} N_{g(y)}^{N_y}. \quad (3.34)$$

A bijective source network exists exactly when g is bijective and $N_y = N_{g(y)}$ for every y . In that case the number of bijective source networks is $\prod_y N_y!$.

Proof. The identity $hf = gh$ is equivalent to $f(x) \in F_{g(y)}$ whenever $x \in F_y$, which is precisely equation (3.33). There are $N_{g(y)}$ choices for the image of each of the N_y states in F_y . This proves the count. All geometric conditions concern the fixed map h , so they impose no further restrictions on these choices.

If f is bijective, g is surjective by $gh = hf$, and therefore bijective. Each target fiber has a unique predecessor fiber. Surjectivity and injectivity of f then force its restriction $F_y \rightarrow F_{g(y)}$ to be a bijection, so the two cardinalities agree. Conversely, if g is bijective and the cardinalities agree, choose a bijection between each such pair of fibers. Their union is a bijective source update. There are $N_y!$ choices on F_y , independently. $\square$

The fiber sizes can be read from the aggregation rules:

$$N_y = \prod_{a=1}^m |H_a^{-1}(y_a)|.$$

In particular, the constraints on bijective lifts depend on the sizes of the observed state fibers as well as on the target dynamics.

Example 3.7.3. Let $h(y, z) = y$ be projection from $m + r$ nodes to m nodes. Every source lift of g has the form

$$f(y, z) = (g(y), F(y, z)),$$

where $F: \mathbb{F}_2^{m+r} \rightarrow \mathbb{F}_2^r$ is arbitrary. The state fibers all have size 2^r , giving $(2^r)^{2^{m+r}}$ possible source networks. The hidden coordinates may depend on both the observed coordinates and one another. Their behavior is not determined by g and h .

Ignored inputs also need not form a subsystem that can simply be deleted. For example,

$$h(x_1, x_2, z) = x_1 + x_2, \quad f(x_1, x_2, z) = (z, z, 0), \quad g(t) = 0$$

define a morphism. Although h ignores z , projection onto (x_1, x_2) is not a semiconjugacy to any two-node network: its next value (z, z) depends on the discarded coordinate. Cancellation makes the scalar observation autonomous.

3.7.3 A realization with the fewest wiring edges

Theorem 3.7.4. Fix a compatible surjection h , a target network g , and an isometric section s as in Proposition 3.3.9. Define

$$f_0 = s \circ g \circ h. \quad (3.35)$$

Then $h: f_0 \rightarrow g$ is a morphism, and

$$|E(W(f_0))| = \sum_{a=1}^m |S_a| d_g^+(a). \quad (3.36)$$

Thus this source network attains the lower bound in Corollary 3.4.3. Its image lies in $s(\mathbb{F}_2^m)$, its restriction to that set is conjugate to g , and $L(f_0) = L(g)$.

Proof. Since $hs = \text{Id}$, we have $hf_0 = gh$. In each block B_b , the section s varies only in the chosen coordinate i_b , which is equal to y_b or its complement. Every other coordinate of s is constant. Consequently, the only possibly nonconstant source update in B_b is

$$(f_0)_{i_b} = \epsilon_b + g_b \circ h \quad \text{for some } \epsilon_b \in \mathbb{F}_2.$$

Corollary 3.3.5 gives

$$\text{Ess}((f_0)_{i_b}) = \bigcup_{\{a: a \rightarrow b \text{ in } W(g)\}} S_a.$$

Counting these edges over b proves equation (3.36). Every other source update is constant, so there are no additional edges.

The image condition follows from the definition, and $f_0 s = sg$ proves conjugacy on the invariant set $s(\mathbb{F}_2^m)$. Since s is isometric and h is a contraction, $L(f_0) \leq L(g)$. Proposition 3.5.5 gives the opposite inequality. $\square$

Every state of this realization reaches $s(\mathbb{F}_2^m)$ after one step. Its periodic orbits correspond bijectively to those of g , with the same lengths. Every directed wiring cycle lies among the selected nodes $i_1, \dots, i_m$, since all other source nodes have constant updates and hence no incoming edges. The wiring subgraph on the selected nodes is isomorphic to $W(g)$.

Equations (3.33) and (3.35) describe two complementary aspects of a reduction. The first enumerates every source dynamics consistent with the observation. The second supplies a source realization attaining the necessary edge and amplification bounds. The derivative identities and lifting theorems above hold throughout this entire family.

Concluding remarks

Bibliography

- [1] Claude Carlet. *Boolean Functions for Cryptography and Coding Theory*. Cambridge University Press, 2021.
- [2] S. De Winter and M. Korb. Weak isometries of the boolean cube. *Discrete Mathematics*, 339(2):877–885, 2016.
- [3] Lee DeVille and Eugene Lerman. Dynamics on networks of manifolds. *Symmetry, Integrability and Geometry: Methods and Applications*, 11:022, 2015. 21 pages.
- [4] Claus Kadelka, Matthew Wheeler, Alan Veliz-Cuba, David Murrugarra, and Reinhard Laubenbacher. Modularity of biological systems: a link between structure and function. *Journal of the Royal Society Interface*, 20(207):20230505, 2023.
- [5] Reinhard Laubenbacher and Bodo Pareigis. Equivalence relations on finite dynamical systems. *Advances in applied mathematics*, 26(3):237–251, 2001.
- [6] Reinhard Laubenbacher and Bodo Pareigis. Decomposition and simulation of sequential dynamical systems. *Advances in Applied Mathematics*, 30(4):655–678, 2003.
- [7] Aymeric Picard Marchetto and Adrien Richard. Isomorphic boolean networks and dense interaction graphs, 2021.
- [8] Christian M Reidys. On certain morphisms of sequential dynamical systems. *Discrete mathematics*, 296(2-3):245–257, 2005.
- [9] Adrien Richard. Local negative circuits and fixed points in non-expansive Boolean networks. *Discrete Applied Mathematics*, 159(11):1085–1093, 2011.
- [10] André Thayse. *Boolean Calculus of Differences*, volume 101 of *Lecture Notes in Computer Science*. Springer, 1981.